

Programme : Cloud Computing and Big Data	
Course Code : 6271C	Course Title: Cloud Security
Semester : 6	Credits: 4
Course Category: Programme Elective Course	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To impart the basics of cloud security.
- To discuss various cloud computing risk issues and security challenges.
- To familiarize with Identity and Access Management (IAM)

Course Prerequisites:

Topic	Course Code	Course name	Semester
Cloud Computing Fundamentals	3273	Introduction to Cloud Computing	3
Basics of computer networks and protocols	4132	Computer Communication and Networks	4
Concepts of advanced cloud computing		Advanced Cloud Computing	5

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the basic concepts related to cloud security.	12	Understanding
CO2	Outline cloud computing risk issues	17	Understanding

CO3	Describe cloud computing security challenges	15	Understanding
CO4	Explain Identity and access Management	14	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						2
CO2	3						2
CO3	3						2
CO4	3						2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the basic concepts related to cloud security.		
M1.01	State Cloud Information Security Objectives	1	Remembering
M1.02	Explain CIA Triads	1	Understanding
M1.03	Explain Cloud Security Services	1	Understanding
M1.04	Describe various Cloud Security Design Principles	3	Understanding
M1.05	Outline Secure Cloud Software Requirements -Secure Development Practices	3	Understanding

M1.06	Summarize Secure Cloud Software Testing	3	Understanding
	Series Test -1	1	
Contents Cloud information security objectives, CIA Triads - Confidentiality, Integrity, and Availability Cloud security services - Authentication, Authorization, Auditing, and Accountability. Cloud security design principles - Least privilege, Separation of duties, Defense in depth Fail safe, Economy of mechanism, Complete mediation, Open design, Least common mechanism, Psychological acceptability, Weakest link, Leveraging existing components Secure Cloud Software Requirements- Secure Development Practices -Handling Data, Code Practices, Language Options, Input Validation, Content Injection and Physical Security of the System Summarize Secure Cloud Software Testing -penetration testing, functional testing, performance testing, and vulnerability testing (General outline only)			
CO2	Outline Cloud Computing Risk issues		
M2.01	Outline common threats to Infrastructure, Data, and Access Control	6	Understanding
M2.02	Describe Cloud Access Control Issues and Database Integrity Issues,Data tampering,Data leakages	2	Understanding
M2.03	Explain Cloud Service Provider Risks	4	Understanding
M2.04	Summarize Key Privacy Concerns in the Cloud	5	Understanding
Threats to Infrastructure, Data, and Access Control, Common Threats and Vulnerabilities - Logon Abuse, Inappropriate System Use, Eavesdropping, Network Intrusion, Denial-of-Service (DoS) Attacks, Session Hijacking Attacks, Fragmentation Attacks. Cloud Access Control Issues, Database Integrity Issues,Data tampering,Data leakage(Refer R1) Cloud Service Provider Risks ,Risk to Virtualized Systems,Back-Door,Spoofing, Man-in-the-Middle, Replay, TCP Hijacking, Social Engineering, Dumpster Diving, Password Guessing, Trojan Horses and Malware. Privacy, Data Life Cycle, Key Privacy Concerns in the Cloud- Access, Compliance, Storage, Retention, Destruction, Audit and monitoring, Privacy breaches, Changes to Privacy Risk Management and Compliance in Relation to Cloud Computing-Collection Limitation Principle, Use Limitation Principle, Security Principle, Retention and Destruction Principle, Transfer Principle, Accountability Principle			

CO3	Describe Cloud Computing Security Challenges		
M3.01	Outline the Security Policy, Implementation and Policy Types	3	Understanding
M3.02	Summarize Virtualization Security Management	1	Understanding
M3.03	Explain Virtual Threats	4	Understanding
M3.04	Describe VM Security Recommendations	4	Understanding
M3.05	Explain VM-Specific Security Techniques	3	Understanding
Contents Security Policy, Implementation, Policy Types- Senior Management Statement of Policy, Regulatory Policies, Advisory Policies, Informative Policies Computer Security Incident Response Team (CSIRT) Virtualization Security Management - Virtualization types, Virtualization management roles Virtual Threats- Vulnerabilities, Hypervisor Risks, Increased Denial of Service Risk VM Security Recommendations - Best Practice Security Techniques- Hardening the Host Operating System, Limiting Physical Access to the Host, Using Encrypted Communications, Disabling Background Tasks, Updating and Patching, Enabling Perimeter Defense on the VM, Implementing File Integrity Checks, Maintaining Backups VM-Specific Security Techniques- Hardening the Virtual Machine, Securing VM Remote Access			
CO4	Explain Identity Management Access Control		
M4.01	Explain IAM Architecture	2	Understanding
M4.02	Illustrate the Identity Life Cycle Management Phases	1	Understanding
M4.03	Summarize the IAM Standards and Protocols for Cloud Services	4	Understanding
M4.04	Explain Identity Management	3	Understanding

M4.05	Outline Access Control	3	Understanding
M4.06	Describe Autonomic Security	1	Understanding
	Series Test – II	1	

Contents

IAM Definitions, IAM Architecture and Practice

Illustrate the Identity Life Cycle Management Phases and operational activities: -

Provisioning, Credential and attribute management, Entitlement management, Compliance management, Identity federation management, Centralization of authentication (authN) and authorization (authZ)

Relevant IAM Standards and Protocols for Cloud Services- IAM Standards and Specifications for Organizations- Security Assertion Markup Language (SAML), Service Provisioning Markup Language (SPML), eXtensible Access Control Markup Language (XACML). IAM Standards, Protocols, and Specifications for Consumers- OpenID, Information cards, Open Authentication (OATH), Open Authentication API (OpenAuth). Identity Management - Passwords, Tokens- Memory Cards and Smart Cards, Biometrics, Implementing Identity Management.

Access Control- Models for Controlling Access, Single Sign-On (SSO).

Autonomic Security, Autonomic Systems, Autonomic Protection, Autonomic Self-Healing

Text / Reference

T/R	Book Title/Author
T1	Cloud Security, Ronald L. Krutz, Russell Dean Vines, Wiley
T2	Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance, Tim Mather, Subra Kumaraswamy, Shahed Latif, O'Reilly Media; 1st edition
R1	B. Alouffi, M. Hasnain, A. Alharbi, W. Alosaimi, H. Alyami and M. Ayaz, "A Systematic Literature Review on Cloud Computing Security: Threats and Mitigation Strategies," in IEEE Access, vol. 9, pp. 57792-57807, 2021, doi: 10.1109/ACCESS.2021.3073203.
R2	Cloud Computing, A Practical Approach, Toby Velte, Anthony Velte, Robert Elsenpeter, Tata McGraw-Hill Osborne Media; 1st edition, 2009
R3	Enterprise Cloud Computing Technology Architecture Applications, Gautam Shroff, Cambridge University Press; 1st edition, 2010

Online Resources

Sl.No	Website Link
1	https://www.ibm.com/cloud-security
2	https://aws.amazon.com/security/
3	https://www.udemy.com/course/cloud-security-architecture-an-introduction/
4	https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9404177
5	https://www.vmware.com/topics/glossary/content/cloud-security.html